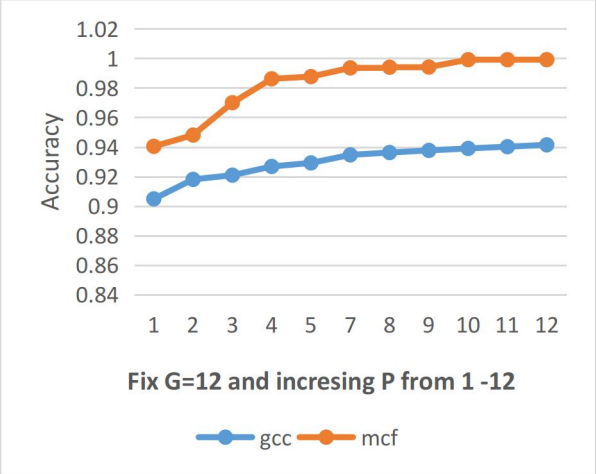
GShare Predictor

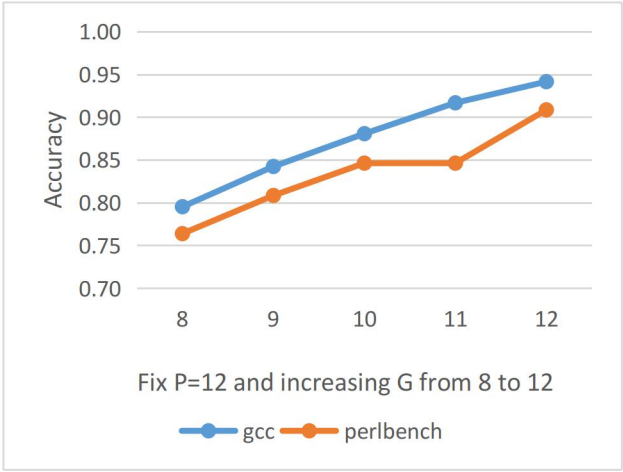
The only relevant parameter for GShare predictor is G, which determines the length of GHR and the number entries in the pattern table. The only valid range considering the size limit is [12, 14], inclusive. The observed pattern is very clear in all traces. The prediction accuracy increases with increasing G. We believe that this is due to that larger G has a greater ability to capture patterns in the program.

Trace / accuracy	G=12	G=13	G=14
gcc	0.930593	0.906406	0.874916
mcf	0.981488	0.981488	0.981495
perlbench	0.832152	0.866587	0.894021
x264	0.881464	0.903848	0.920462
Average	0.90642425	0.91458225	0.9177235

Yeh-Patt Predictor

For Yeh-Patt predictor, there are two parameters influencing the prediction result. We choose to fix one of the parameter, and observe the impact of changing the other parameter. The first figure shows the result for two traces running with G=12 and increasing P from 1 to 12. The second figures shows two traces running with P=12 and increasing G from 8 to 12.





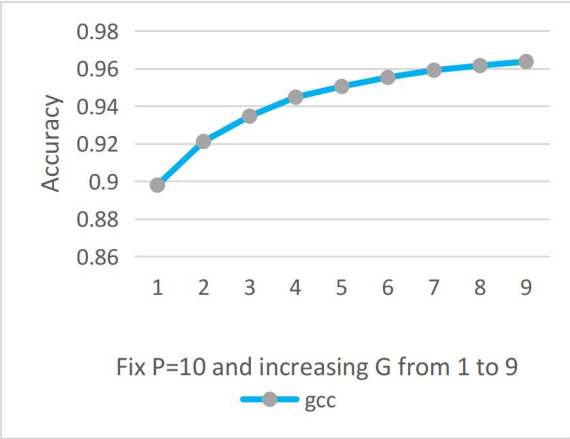
Based on the above, we should be able to conclude that both increasing G and P will improve the predictor performance. Therefore, the most accurate configuration for Yeh-Patt predictor should be either G=14, P=11, or G=12, p=12, considering the size limit.

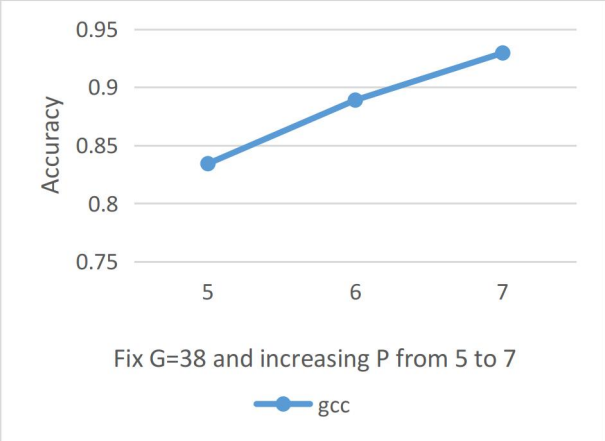
	GCC	MCF	Perlbench	X264	Average accuracy
G=14, P=11	0.960152	0.999139	0.935302	0.924325	0.9547295
G=12, P=12	0.941419	0.999132	0.908492	0.921143	0.9425465

In conclusion, the best Yeh-Patt predictor is G=14, P=11, with average accuracy 0.9547295.

Perceptron Predictor

The relevant parameters for perceptron predictors are P and G. Since we cannot fix a single P value and have it cover the entire G range, only a subset of G values is covered. But, it should be sufficient for us to see the trend. Same scenario happens with fixing G.





Based on the above, both increasing G and P help improving the accuracy. Therefore, the best perceptron should be the one maximizing P and G given the total size limit. We can work out all possible configurations of P and G and run those to see performance.

G	P	gcc	mcf	perlbench	x264	Average accuracy
63	5	0.845256	0.993257	0.796288	0.879849	0.8786625
63	6	0.898382	0.994314	0.847604	0.925316	0.916404
63	7	0.936807	0.994337	0.890876	0.944968	0.941747
35	8	0.956316	0.994289	0.919927	0.948859	0.95484775
20	**9**	**0.966744**	**0.981723**	**0.93747**	**0.948499**	**0.958609**
9	10	0.963747	0.976885	0.936194	0.937763	0.95364725
5	11	0.959833	0.976999	0.936269	0.928357	0.9503645
2	12	0.950919	0.97178	0.922958	0.90998	0.93890925
1	13	0.949532	0.97178	0.925995	0.90799	0.93882425

Conclusion

Overall, with an average accuracy of 0.95806, the perceptron predictor with G=20 and P=9 is the best branch predictor with regards to the given traces.